

Part 1: Creating AI optimization models and making sense of their output

In your favorite chatbot, do the following: (1) upload a copy of the PDF with the case and (2) copy-paste the following structured prompt:

Please build an optimization model for me with the following specs:

- *maximize the gross profit for the quarter*
- *meet all the physical constraints (e.g., capacities at factories and centers)*
- *for any client, do not ship more than the client pre-orders*
- *no requirements are imposed on the fill rates achieved at each client.*

Based on the model's recommendations, please display for me the following information:

- *the number of units produced in each factory, processed in each center, and shipped to each client*
- *each client's fill rate and my resulting gross profit*

Please give me code that I can run without requiring any licenses. Ideally, you should run the code for me and display everything. If you cannot do that, give me code that is self-contained (so I can just copy-paste it and run it somewhere) and explain to me where I can copy-paste it and run it.

In the AI cannot run your code or does not give you good hints for where to run it, navigate to <https://colab.research.google.com/> in your browser and prompt the instructor for help. If the code still does not run, just look at the output below!

Compare the output from the chatbot with the output below and address the following questions:

Q1. Does your chatbot's recommendation match the one below? Do these decisions make "practical" sense?

Do the specs we used above make sense? (brainstorming)

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Solved the optimization AI Model. Here is the recommendation:
Gross profit: 33,625,000.
Number of units produced at the factories:
  Zuffe : 5,000 (capacity was: 5,000)
  Leip  : 0      (capacity was: 2,000)
Number of units shipped from factories to customization centers:
  Zuffe to C1 : 2,000
  Zuffe to C2 : 2,200
  Zuffe to C3 : 800
  Leip  to C1 : 0
  Leip  to C2 : 0
  Leip  to C3 : 0
Number of units processed at customization centers:
  C1 : 2,000 (capacity was: 2,000)
  C2 : 2,200 (capacity was: 2,200)
  C3 : 800   (capacity was: 3,300)
Number of units shipped from customization centers to clients:
  C1 to OO : 100
  C1 to FL : 1,900
  C1 to SH : 0
  C1 to WW : 0
  C2 to OO : 0
  C2 to FL : 600
  C2 to SH : 1,600
  C2 to WW : 0
  C3 to OO : 250
  C3 to FL : 0
  C3 to SH : 0
  C3 to WW : 550
Filled orders and fill rates at clients:
  OO : 350 orders filled out of 5,000, i.e., fill rate of 7.0 %
  FL : 2,500 orders filled out of 2,500, i.e., fill rate of 100.0 %
  SH : 1,600 orders filled out of 1,600, i.e., fill rate of 100.0 %
  WW : 550 orders filled out of 550 , i.e., fill rate of 100.0 %
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*This mini-case was prepared by Professor Dan Iancu (daniiancu@stanford.edu) as a basis for class discussion. It is not intended to serve as an endorsement, source of primary data, or illustration of effective or ineffective management.

Part 2: Navigating tradeoffs

In your favorite chatbot, try the following structured prompt (copy-paste):

Now, let's do the following: I want you to create a model with the same specs as the one we just considered, except that for any client, the model should ensure that the fill rate exceeds a required level f . I want you to change the required fill rate f and determine the best plan for every required fill rate. And please help me visualize how the fill rates at the clients and how my gross profit depend on this requirement.

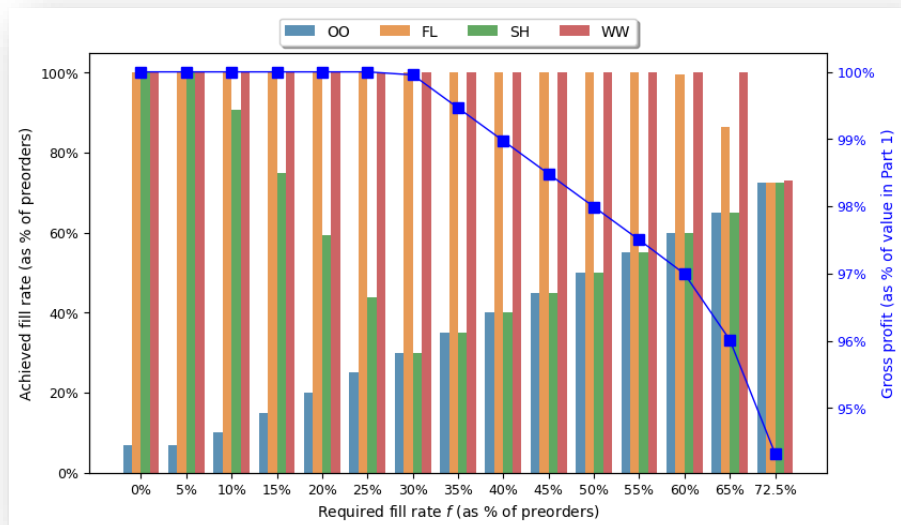
And make sure you give me code that I can run without requiring any licenses. Ideally, you should run the code for me and display everything. If you cannot do that, give me code that is self-contained (i.e., so I can just copy-paste it somewhere) and explain to me where I can copy-paste it to run it as is.

Compare the output from the chatbot with the output below and address the following questions:

Q2. How would you set the minimum fill rate requirement? (brainstorming)

Clarification about the figure.

- the **vertical colored bars** correspond to the **left axis** and **the legend**, and show the fill rate actually achieved at each client;
- the **blue line plot** corresponds to the **right axis** and shows the profit achieved, as a percent of the profit reported in Part 1.



Part 3: Pricing bottlenecks

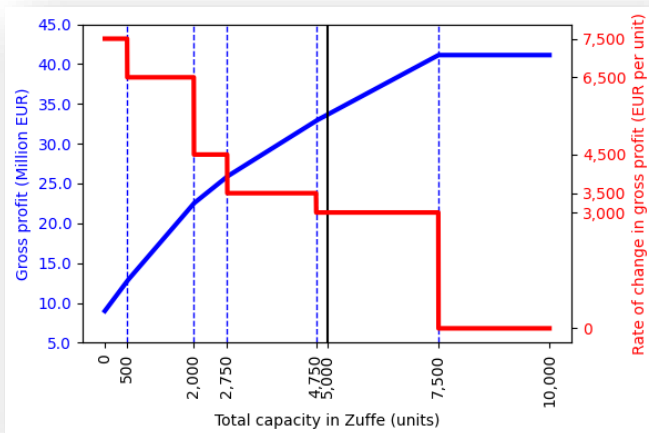
In your favorite chatbot, try the following structured prompt (copy-paste):

I am considering expanding the capacity of the manufacturing facilities and customization centers, but I am not sure where. Can you rerun the initial model we built (without requirements on client fill rates) but change the capacity at each factory, one at a time, and show me how my gross profit depends on each capacity? Based on this, if I were to expand capacity by one unit, where should I expand?

Compare the output from the chatbot with the output below and address the following questions:

Q3. If you were to increase capacity at one facility, which would you choose? (brainstorming)

Brainstorming: Do you use models like these to inform decisions in your organization? If yes, what kind of decisions? If not, what are decisions that could benefit from this and what are implementation barriers?



A few clarifications about each figure.

- The **blue plot** corresponds to the **left axis** and shows the **gross profit**.
- The **dashed blue vertical lines** mark the points where the slope of the blue line changes.
- The **red plot** corresponds to the **right axis** and shows the **rate of change in the profit**, i.e., the slope of the blue line.
- The **solid black vertical line** marks the value of the capacity mentioned in the case for that facility.

